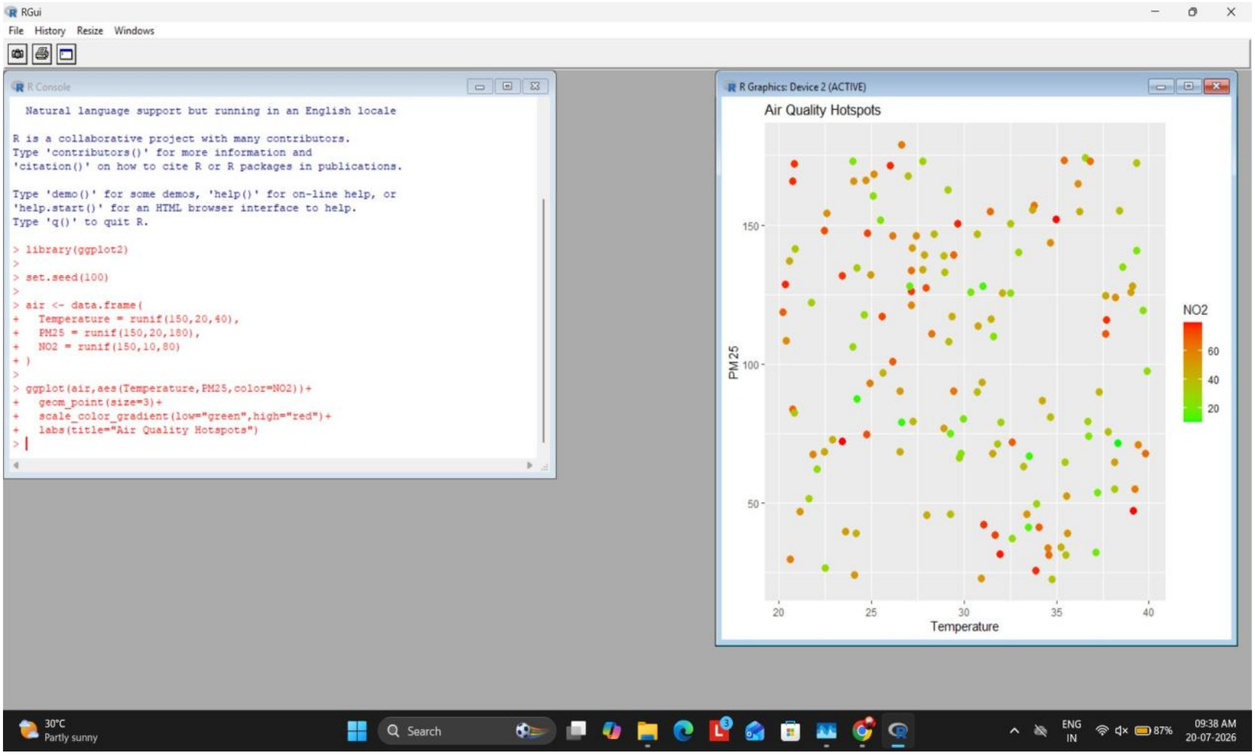
R Programs with Output Screenshots

Question 1

```
library(ggplot2) set.seed(100) air<-
data.frame(Temperature=runif(150,20,40),PM25=runif(150,20,180),NO2=runif(150,
10,80))
ggplot(air,aes(Temperature,PM25,color=NO2))+geom_point(size=3)+scale_color_gr
adient(low='green',high=
```

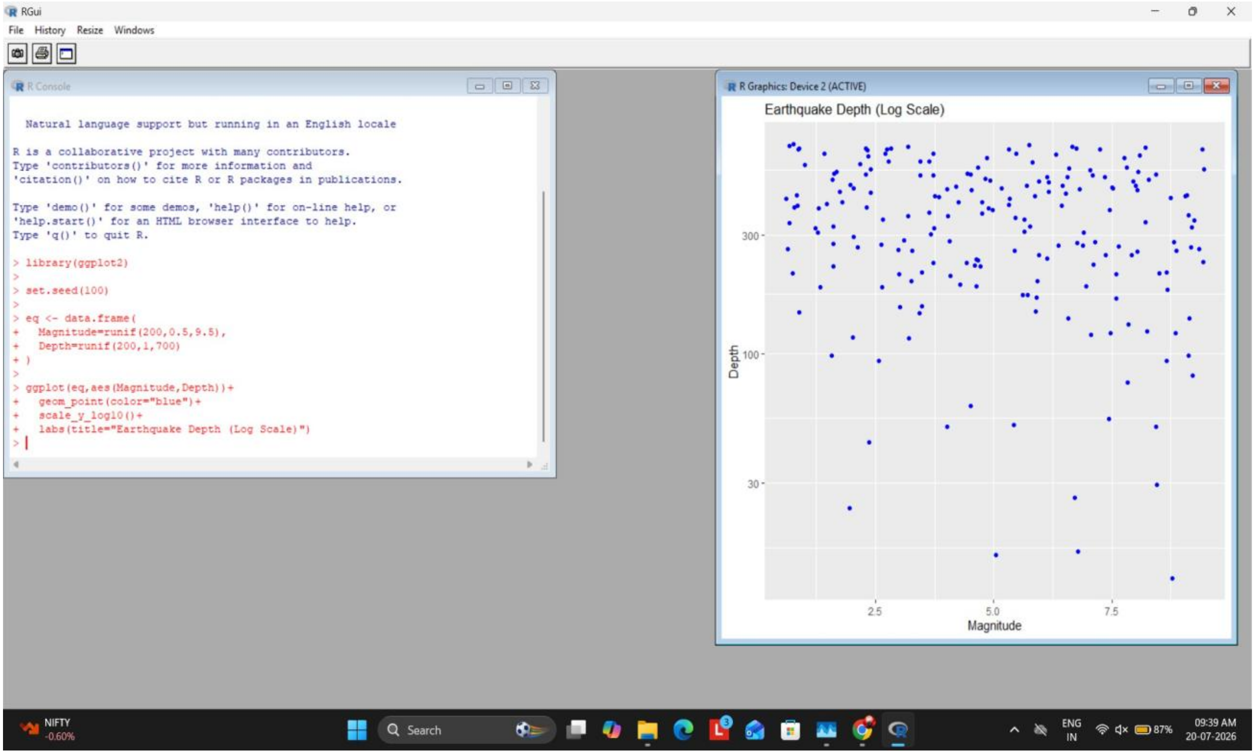
Output Screenshot:



Question 2

```
libra
ry(gg
plot2
)
set.s
eed(1
00)
eq<-
data.frame(Magnitude=runif(200,0.5,9.5),Depth=run
if(200,1,700))
ggplot(eq,aes(Magnitude,Depth))+geom_point(color='blue')+scale_y_log10()+labs(title=
                                                          'Earthquake Depth
```

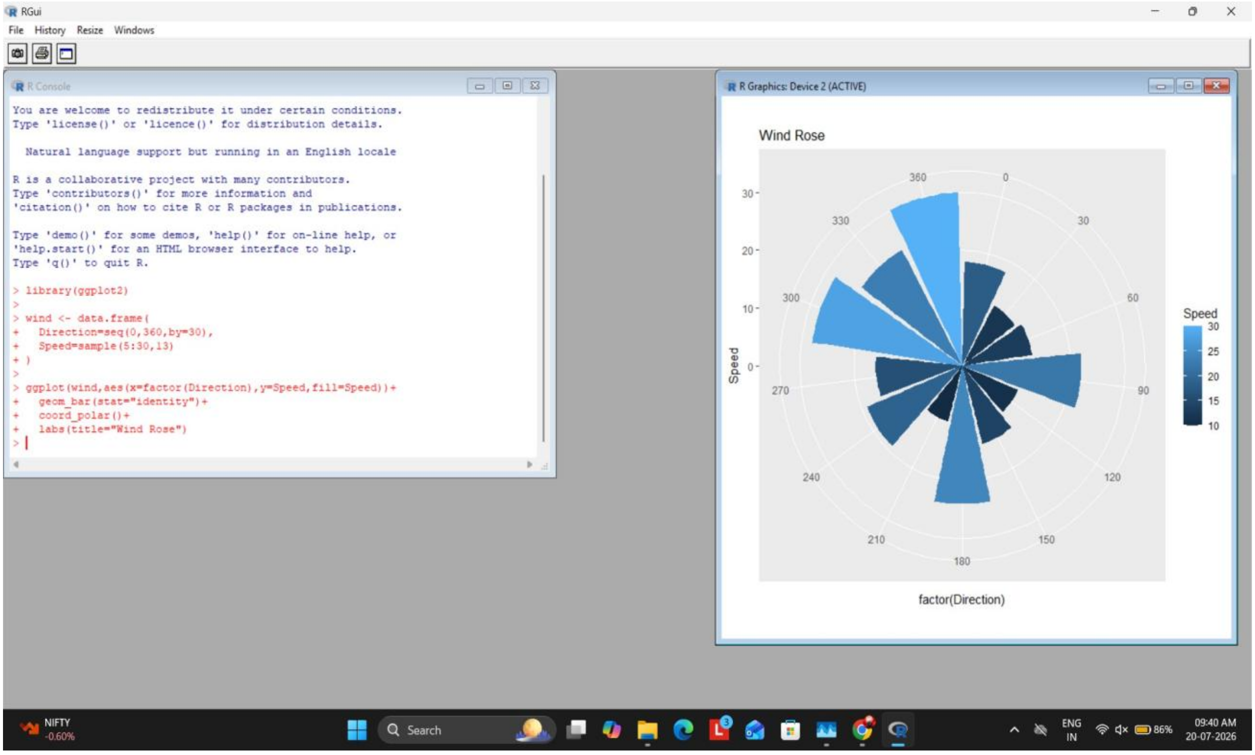
Output Screenshot:



Question 3

```
library(ggplot2) wind<-
data.frame(Direction=seq(0,360,30),Speed=sample(5
:30,13))
ggplot(wind,aes(factor(Direction),Speed,fill=Speed))+geom_bar(stat='identity')+coord
_polar()+labs(tit
```

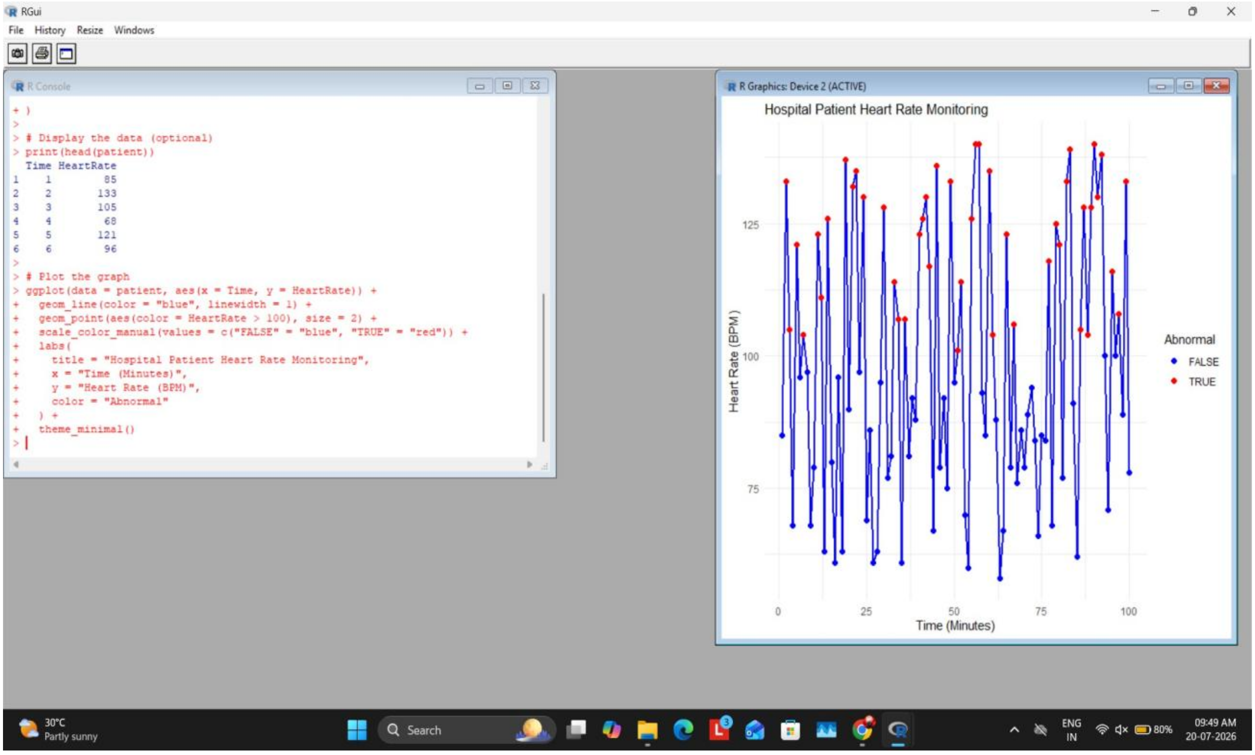
Output Screenshot:



Question 4

```
library(ggplot2) set.seed(123) patient<-
data.frame(Time=1:100,HeartRate=sample(55:140,100
,replace=TRUE))
ggplot(patient,aes(Time,HeartRate))+geom_line(color='blue')+geom_point(aes(color=Hea
rtRate>100))+scal
```

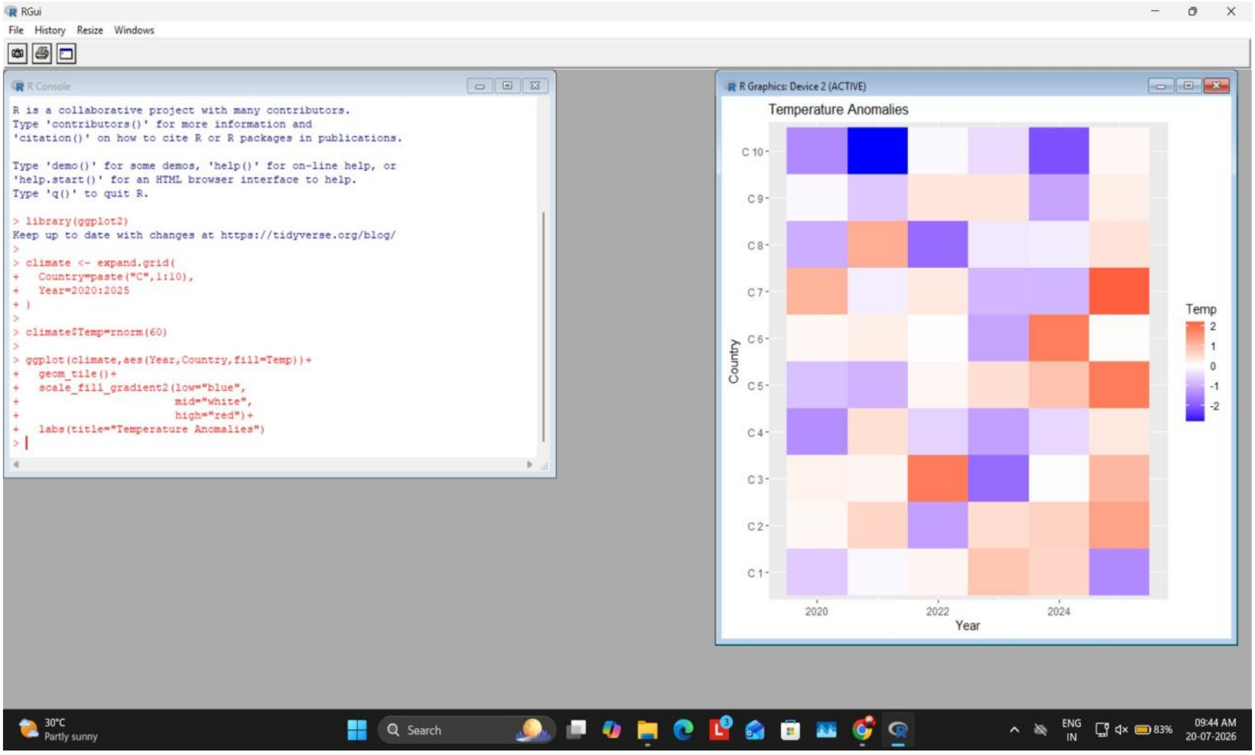
Output Screenshot:



Question 5

```
library(ggplot2) climate<-
expand.grid(Country=paste('C',1:10),Year=2020:202
5) climate$Temp=rnorm(60)
ggplot(climate,aes(Year,Country,fill=Temp))+geom_tile()+scale_fill_gradient2(low='bl
ue',mid='white',h
```

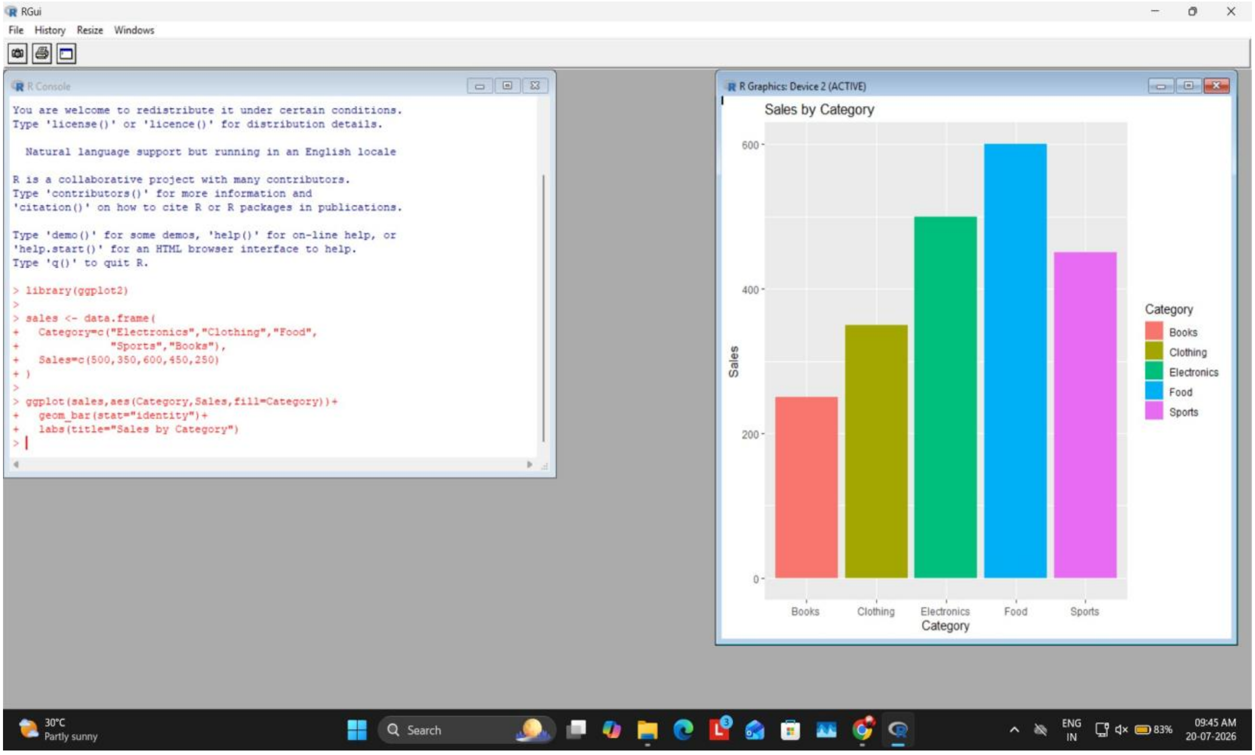
Output Screenshot:



Question 6

```
library(ggplot2) sales<-
data.frame(Category=c('Electronics','Clothing','Food','Sports','Books'),Sales
=c(500,350,600,45
ggplot(sales,aes(Category,Sales,fill=Category))+geom_bar(stat='identity')
```

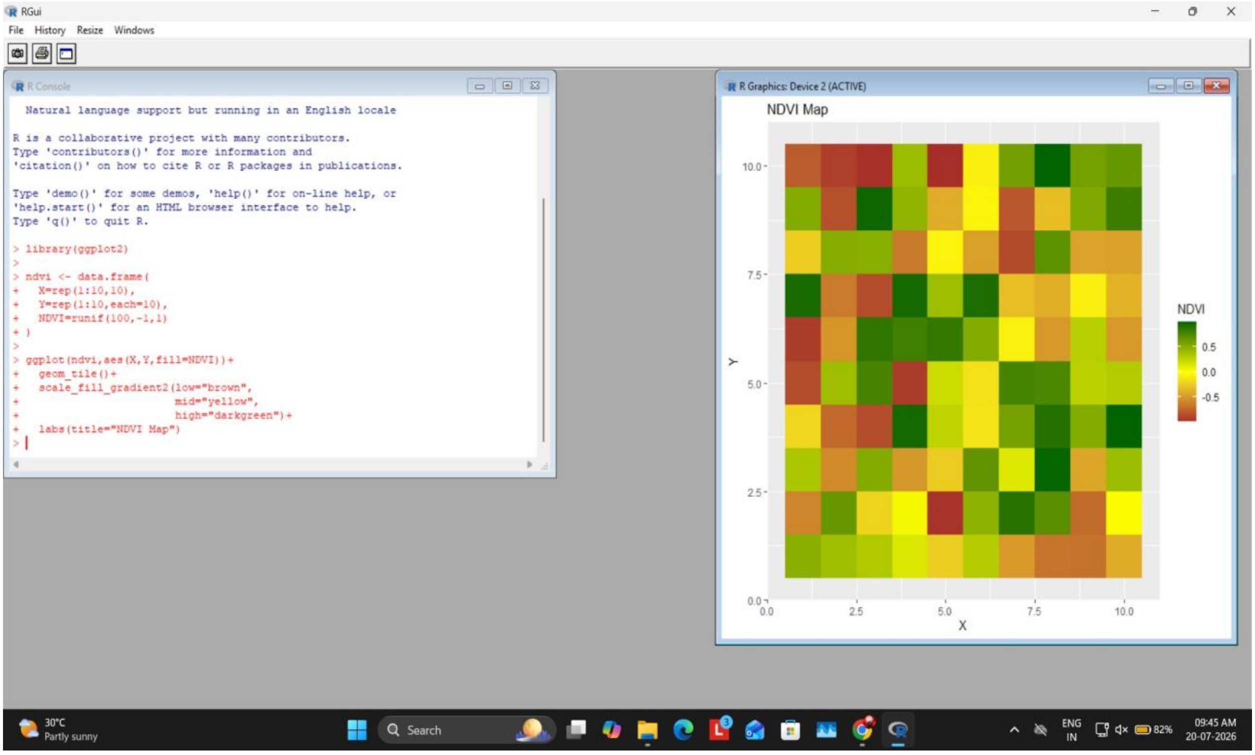
Output Screenshot:



Question 7

```
library(ggplot2) ndvi<-
data.frame(X=rep(1:10,10),Y=rep(1:10,each=10),NDV
I=runif(100,-1,1))
ggplot(ndvi,aes(X,Y,fill=NDVI))+geom_tile()+scale_fill_gradient2(low='brown',mid='ye
llow',high='darkg
```

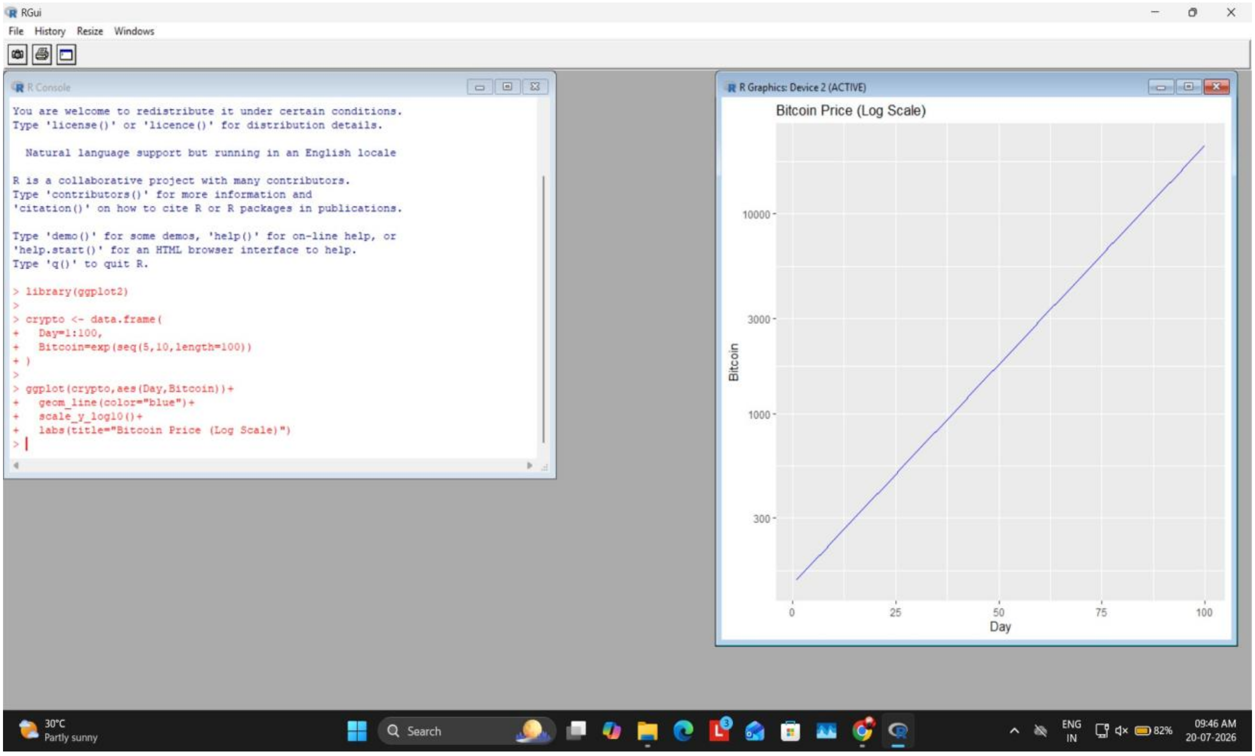
Output Screenshot:



Question 8

```
library(ggplot2)
crypto<-
data.frame(Day=1:100,Bitcoin=exp(seq(5,10,length=
100)))
ggplot(crypto,aes(Day,Bitcoin))+geom_line(color='
blue')+scale_y_log10()
```

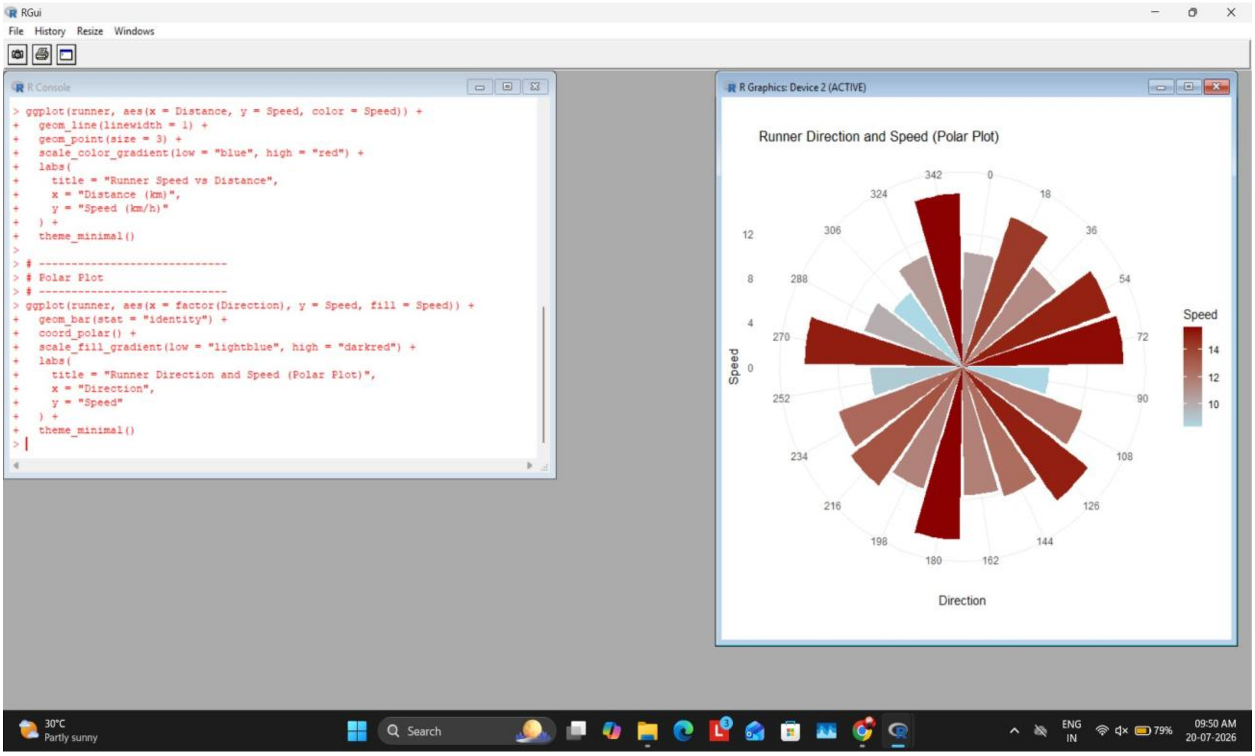
Output Screenshot:



Question 9

```
library(ggplot2) runner<-
data.frame(Distance=1:20,Speed=runif(20,8,16),Direction=seq(0,342,
length.out=20)) # Cartesian and Polar plots
```

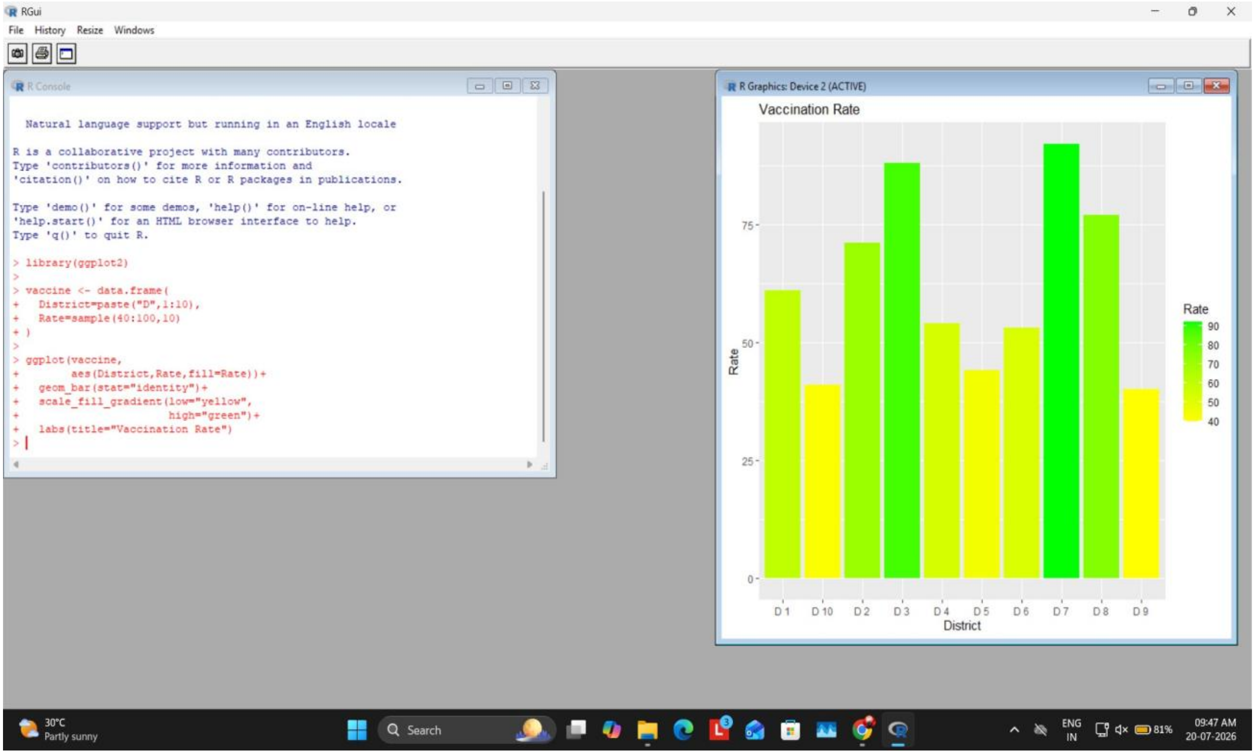
Output Screenshot:



Question 10

```
library(ggplot2) vaccine<-
data.frame(District=paste('D',1:10),Rate=sample(4
0:100,10))
ggplot(vaccine,aes(District,Rate,fill=Rate))+geom_bar(stat='identity')+scale_fill_gr
adient(low='yello
```

Output Screenshot:

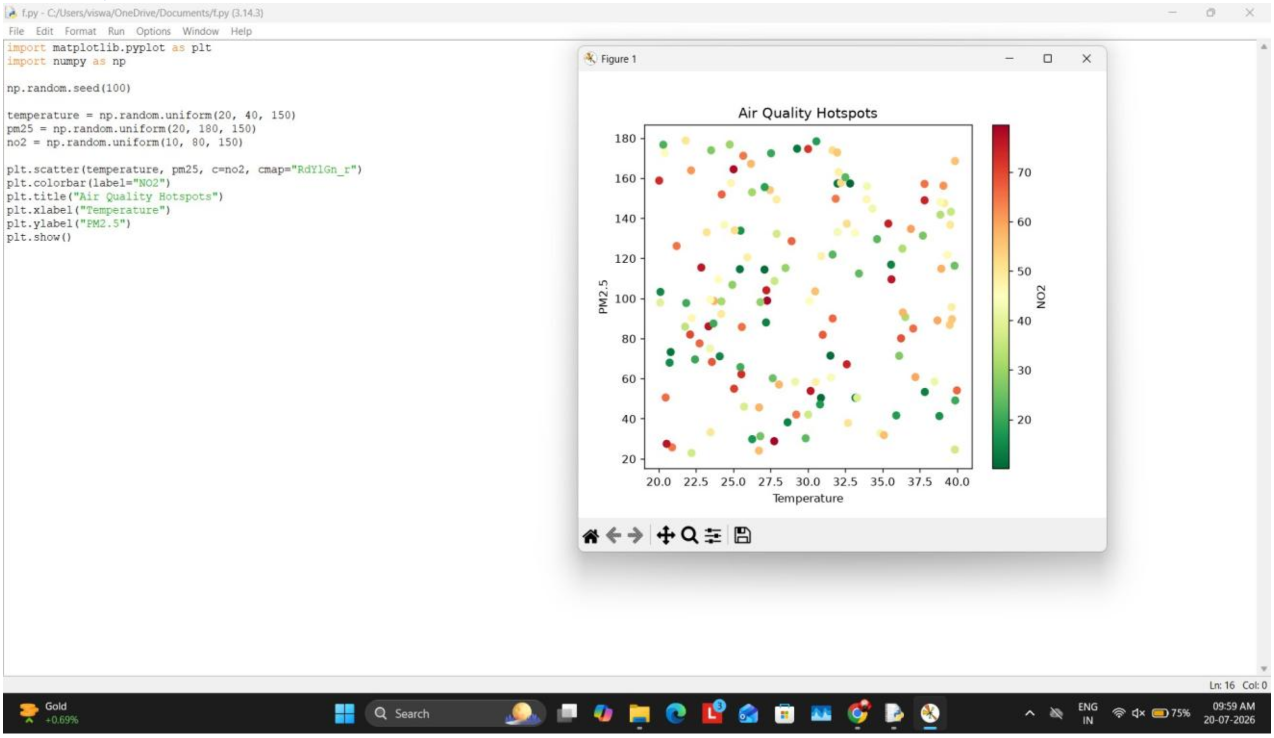


Python Programs with Output Screenshots Question 1

```
import matplotlib.pyplot as plt
import numpy as np
np.random.seed(100)
```

```
temperature = np.random.uniform(20, 40, 150)
pm25 = np.random.uniform(20, 180, 150) no2 =
np.random.uniform(10, 80, 150)
plt.scatter(temperature, pm25, c=no2, cmap="RdYlGn_r")
plt.colorbar(label="NO2") plt.title("Air Quality
Hotspots") plt.xlabel("Temperature")
plt.ylabel("PM2.5") plt.show()
```

Output Screenshot



Question 2

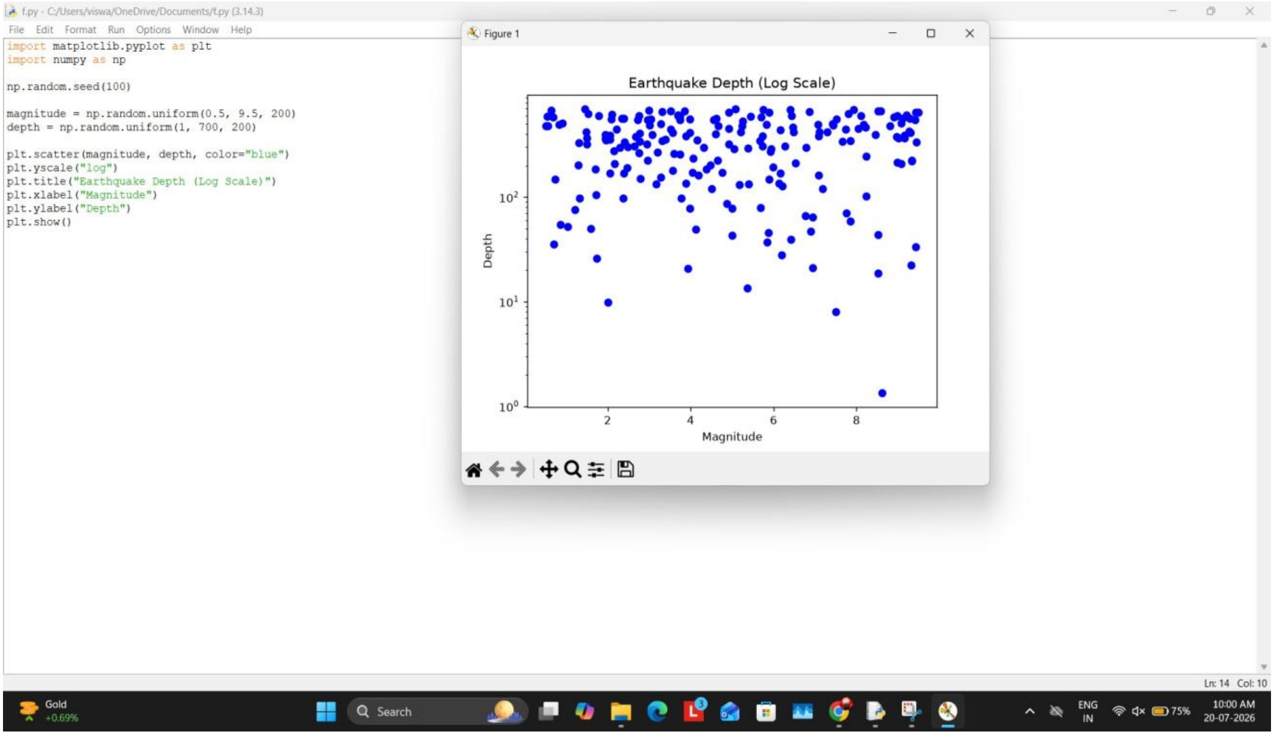
```
import matplotlib.pyplot as plt

import numpy as np

np.random.seed(100)

magnitude = np.random.uniform(0.5, 9.5, 200)
depth = np.random.uniform(1, 700, 200)
plt.scatter(magnitude, depth, color="blue")
plt.yscale("log") plt.title("Earthquake
Depth (Log Scale)") plt.xlabel("Magnitude")
plt.ylabel("Depth") plt.show()
```

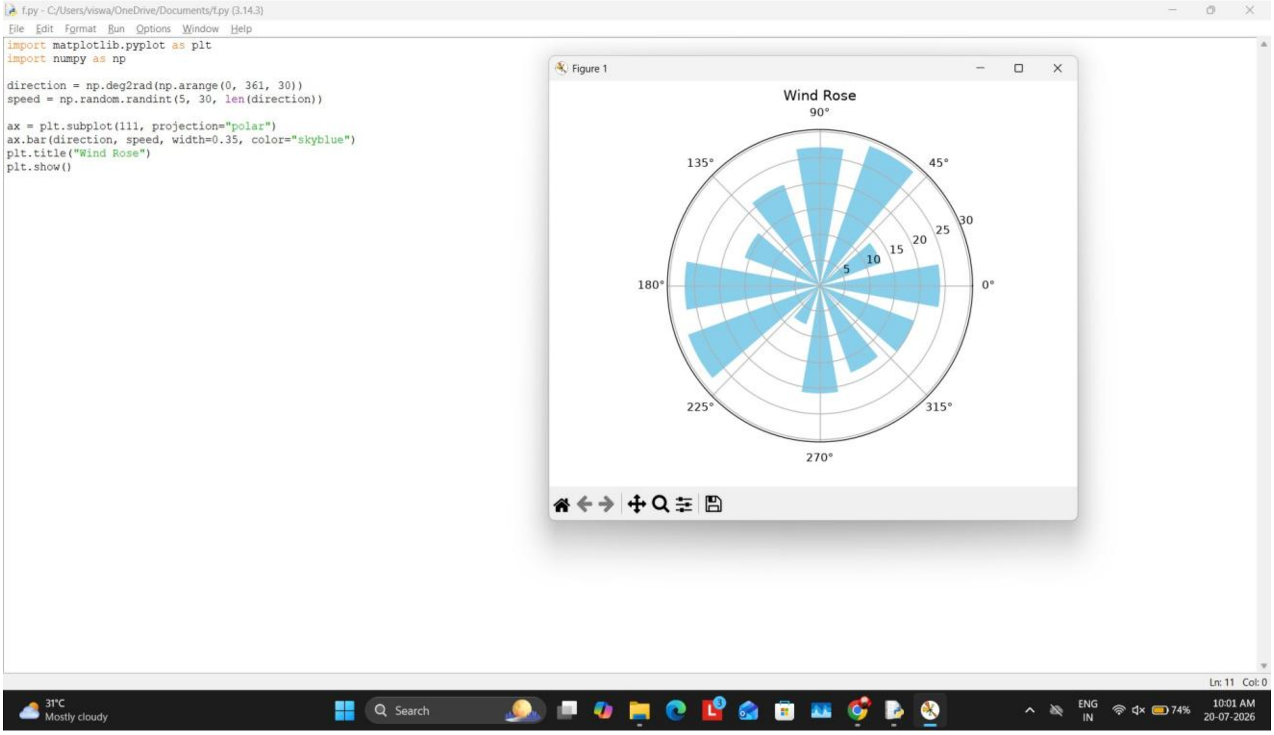
Output Screenshot



Question 3

```
import matplotlib.pyplot as plt
import numpy as np
direction = np.deg2rad(np.arange(0, 361, 30))
speed = np.random.randint(5, 30, len(direction))
ax = plt.subplot(111, projection="polar")
ax.bar(direction, speed, width=0.35, color="skyblue")
plt.title("Wind Rose") plt.show()
```

Output Screenshot



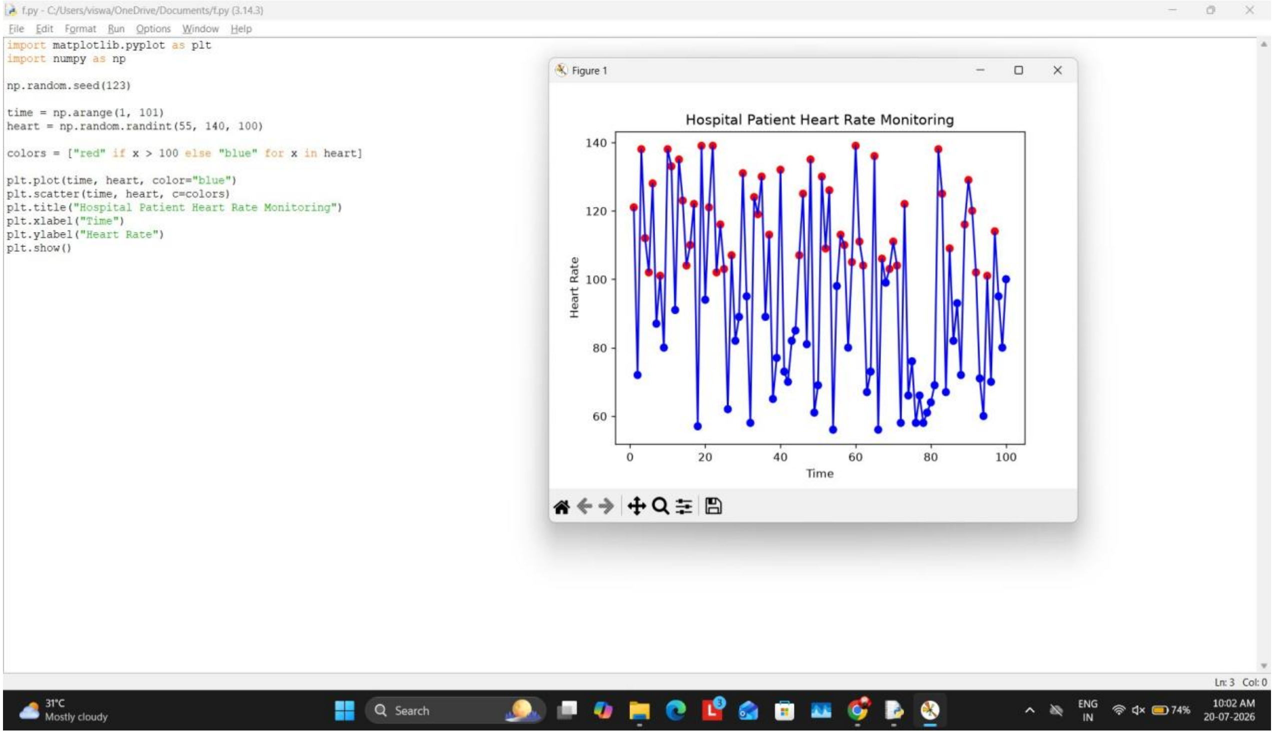
Question 4

```
import matplotlib.pyplot as plt
import numpy as np
np.random.seed(123)

time = np.arange(1, 101) heart = np.random.randint(55,
140, 100) colors = ["red" if x > 100 else "blue" for x
in heart]

plt.plot(time, heart, color="blue")
plt.scatter(time, heart, c=colors)
plt.title("Hospital Patient Heart Rate Monitoring")
plt.xlabel("Time") plt.ylabel("Heart Rate")
plt.show()
```

Output Screenshot

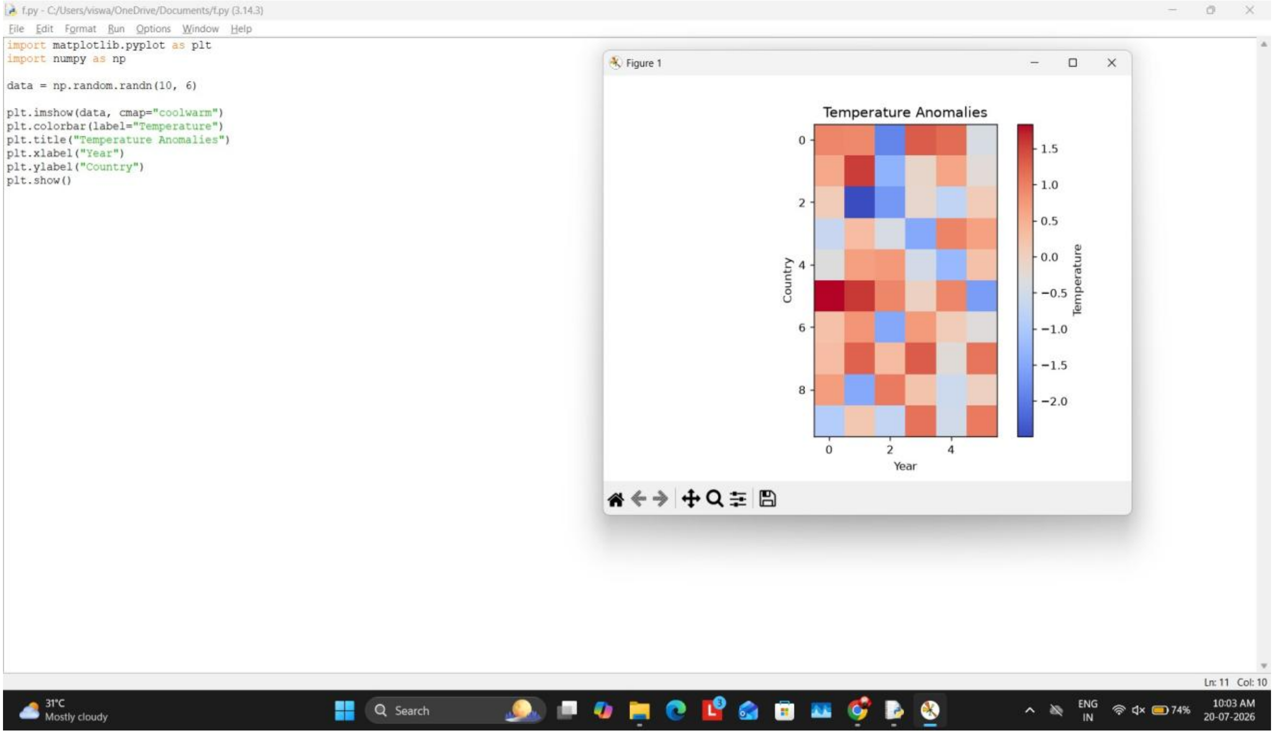


Question 5

```
import matplotlib.pyplot as plt
import numpy as np data =
np.random.randn(10, 6)

plt.imshow(data, cmap="coolwarm")
plt.colorbar(label="Temperature")
plt.title("Temperature Anomalies")
plt.xlabel("Year")
plt.ylabel("Country") plt.show()
```

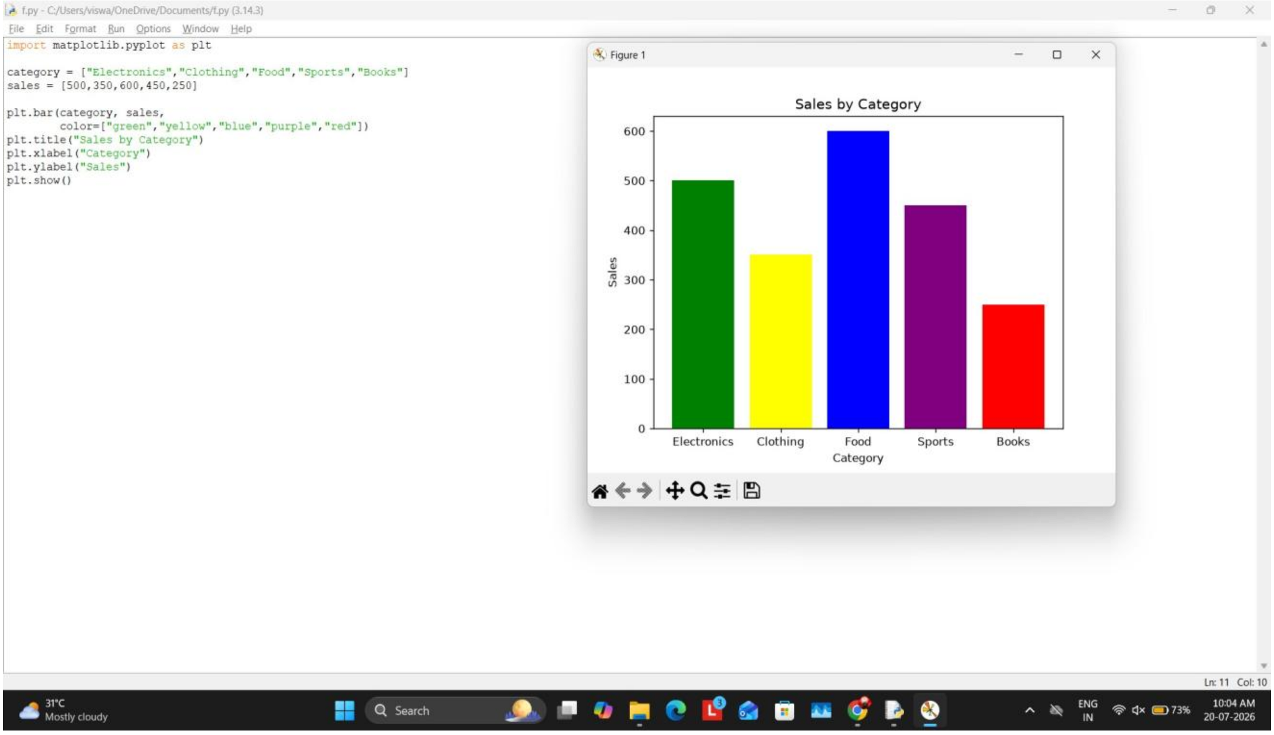
Output Screenshot



Question 6

```
import matplotlib.pyplot as plt
category = ["Electronics","Clothing","Food","Sports","Books"]
sales = [500,350,600,450,250]
plt.bar(category, sales,
color=["green","yellow","blue","purple","red"])
plt.title("Sales by Category") plt.xlabel("Category")
plt.ylabel("Sales") plt.show()
```

Output Screenshot

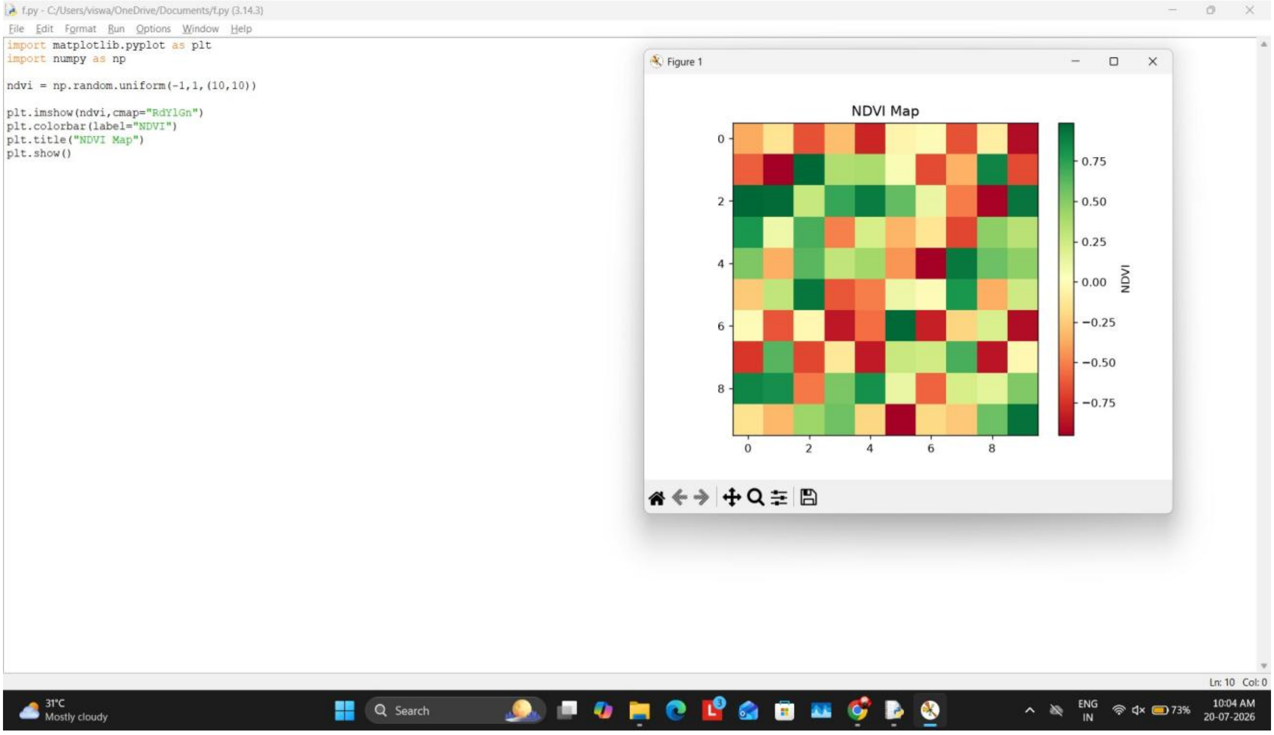


Question 7

```
import matplotlib.pyplot as plt import
numpy as np ndvi = np.random.uniform(-
1,1,(10,10))

plt.imshow(ndvi,cmap="RdYlGn")
plt.colorbar(label="NDVI")
plt.title("NDVI Map")
plt.show()
```

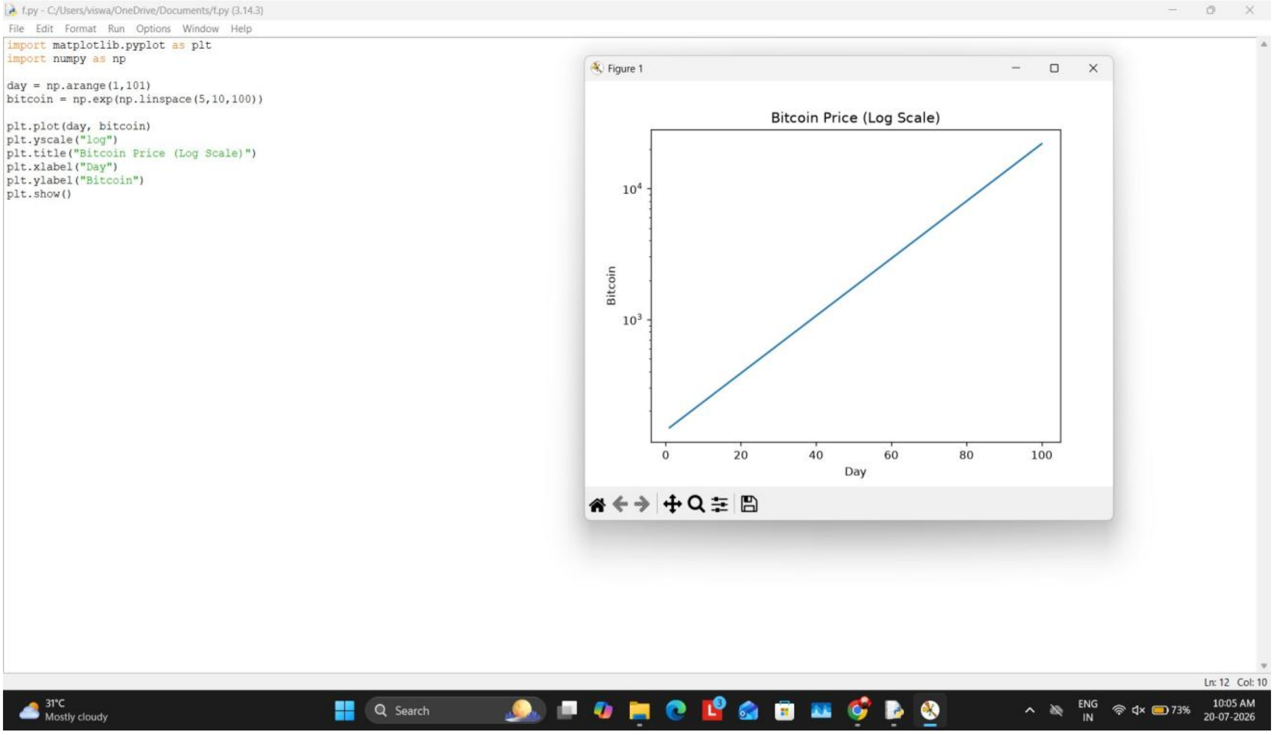
Output Screenshot



Question 8

```
import matplotlib.pyplot as plt
import numpy as np
day = np.arange(1,101) bitcoin =
np.exp(np.linspace(5,10,100))
plt.plot(day, bitcoin)
plt.yscale("log") plt.title("Bitcoin
Price (Log Scale)") plt.xlabel("Day")
plt.ylabel("Bitcoin") plt.show()
```

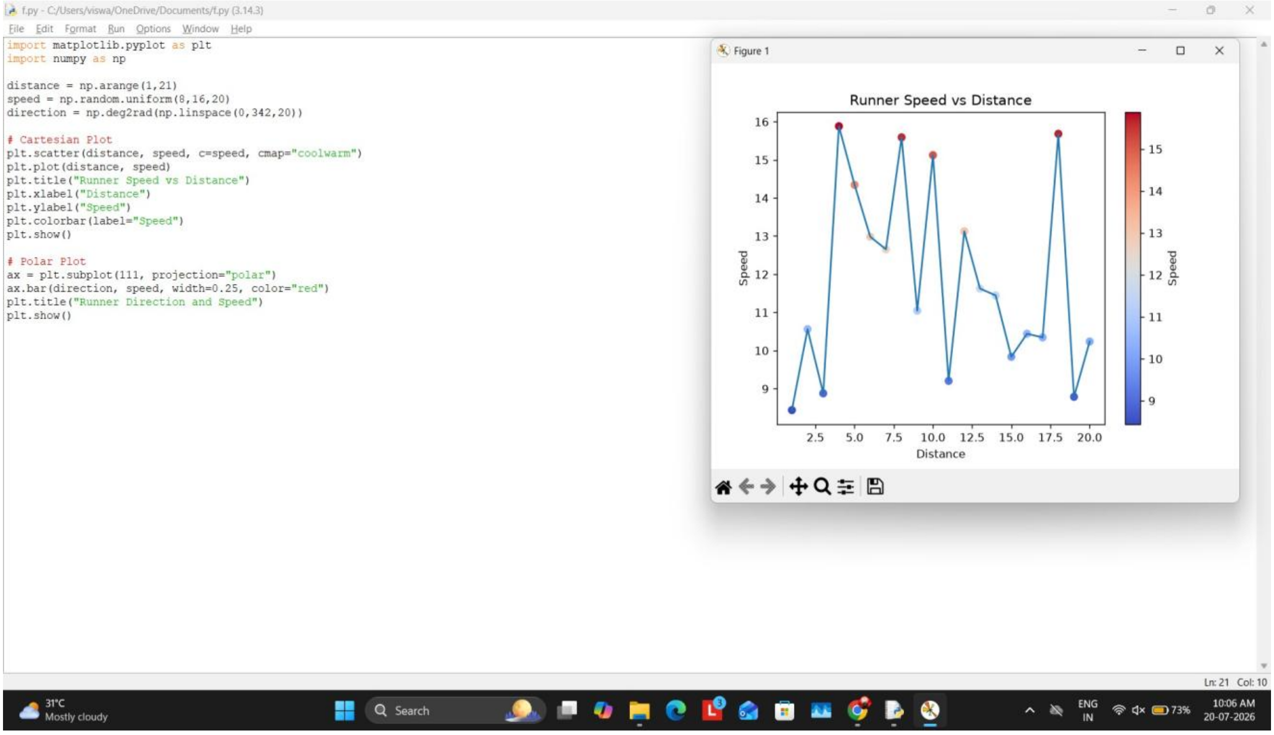
Output Screenshot



Question 9

```
import matplotlib.pyplot as plt
import numpy as np
distance = np.arange(1,21) speed =
np.random.uniform(8,16,20) direction =
np.deg2rad(np.linspace(0,342,20))
# Cartesian Plot plt.scatter(distance, speed, c=speed,
cmap="coolwarm") plt.plot(distance, speed)
plt.title("Runner Speed vs Distance")
plt.xlabel("Distance") plt.ylabel("Speed")
plt.colorbar(label="Speed") plt.show()
# Polar Plot
ax = plt.subplot(111, projection="polar")
ax.bar(direction, speed, width=0.25, color="red")
plt.title("Runner Direction and Speed")
plt.show()
```

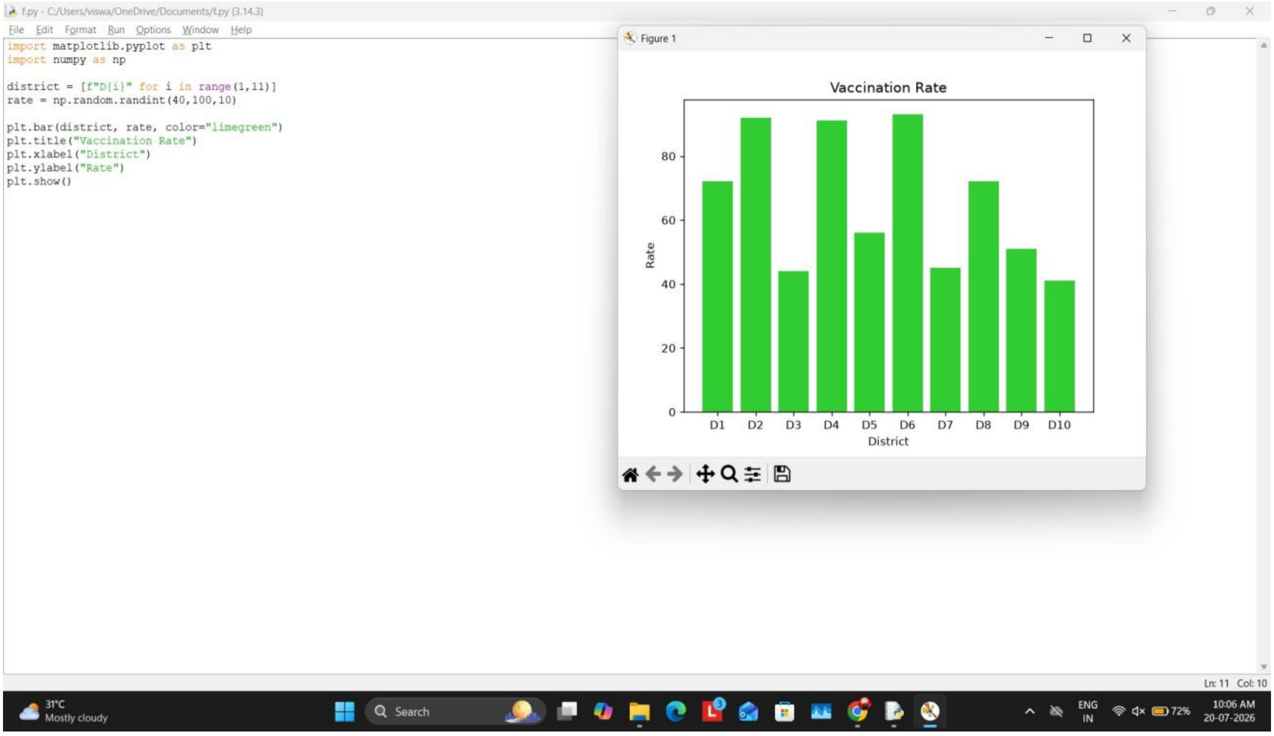
Output Screenshot



Question 10

```
import matplotlib.pyplot as plt
import numpy as np
district = [f"D{i}" for i in range(1,11)]
rate = np.random.randint(40,100,10)
plt.bar(district, rate, color="limegreen")
plt.title("Vaccination Rate")
plt.xlabel("District") plt.ylabel("Rate")
plt.show()
```

Output Screenshot


f.py - C:/Users/viswa/OneDrive/Documents/f.py (3.14.3)
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
import numpy as np
district = [f"D{i}" for i in range(1,11)]
rate = np.random.randint(40,100,10)
plt.bar(district, rate, color="limegreen")
plt.title("Vaccination Rate")
plt.xlabel("District")
plt.ylabel("Rate")
plt.show()
Figure 1
Vaccination Rate
Rate
District
D1 D2 D3 D4 D5 D6 D7 D8 D9 D10
0 20 40 60 80
Ln: 11 Col: 10
31°C
Mostly cloudy
Search
ENG
IN
72%
10:06 AM
20-07-2026